Chapter 4: Monte Carlo and Variational Inference

Part III — Statistical Foundations of Deep Learning & Reinforcement Learning

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Abstract

Most quantities a modern model needs — an expected loss, a posterior mean, a marginal likelihood — are integrals with no closed form. There are two ways to attack an intractable integral, and this chapter is about both. **Monte Carlo** estimates it by sampling: the average of a function over random draws is an unbiased estimate of its expectation, with error that falls like $1/\sqrt{n}$ regardless of dimension, and importance sampling reweights draws from a convenient proposal when the target is hard to sample. **Variational inference** takes the opposite tack: it turns the integral into an *optimisation*, maximising a tractable lower bound — the ELBO — on the log-evidence. That bound is the same one that drove EM in Part II, and it is the objective the variational autoencoder trains.

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1 Monte Carlo estimation

To estimate $\mu = \mathbb{E}_p[f(X)] = \int f(x)p(x)\,dx$, draw $X_1, \dots, X_n \sim p$ and average.

Theorem 1.1 (The Monte Carlo estimator is unbiased, with $1/\sqrt{n}$ error)**.** Let $\hat{\mu}_n = \frac{1}{n}\sum_{i=1}^{n} f(X_i)$ with $X_i \sim p$ i.i.d. and $\mathrm{Var}_p(f(X)) = \sigma^2 < \infty$. Then $\mathbb{E}[\hat{\mu}_n] = \mu$ and $\mathrm{Var}(\hat{\mu}_n) = \sigma^2/n$, so the standard error is $\sigma/\sqrt{n}$.

Proof. Linearity of expectation gives $\mathbb{E}[\hat{\mu}_n] = \mu$, and independence gives

$$\mathrm{Var}(\hat{\mu}_n) = \frac{1}{n^2}\sum_{i} \mathrm{Var}(f(X_i)) = \frac{\sigma^2}{n},$$

whose square root is the standard error. □

The decisive feature is what is *absent*: the rate $1/\sqrt{n}$ does not depend on the dimension of x. A grid quadrature needs k^d points in d dimensions; Monte Carlo needs the same n for a given accuracy whether d is one or a million. That is why it, and not quadrature, is how expectations are computed in deep learning. The left panel shows the estimator of $\mathbb{E}[X^2]$ settling onto the truth inside its shrinking $\pm 1.96\,\sigma/\sqrt{n}$ band.

Definition 1.1 (Importance sampling)**.** When p is hard to sample, draw from a proposal q and reweight:

$$\mu = \mathbb{E}_p[f(X)] = \mathbb{E}_q\Big[f(X)\,\tfrac{p(X)}{q(X)}\Big] \approx \frac{1}{n}\sum_{i=1}^{n} f(X_i)\,\tfrac{p(X_i)}{q(X_i)}, \quad X_i \sim q.$$

The ratio $w(x) = p(x)/q(x)$ is the **importance weight**.

Importance sampling is unbiased for any q with $q > 0$ wherever $fp \neq 0$, but its variance depends sharply on the match between q and fp. Its signature use is rare events: to estimate $P(X > 4)$ for a standard normal, naive Monte Carlo almost never draws a sample past 4 and returns zero, while sampling from a proposal centred out in the tail and reweighting gives a tight estimate — the middle panel, where importance sampling tracks the true 3.2×10^{-5} and naive Monte Carlo is useless.

2 Variational inference: the integral as an optimisation

Sometimes the integral we want is the **evidence** $p(x) = \int p(x, z)\, dz$, the normaliser of a posterior over a latent z. Sampling it well can be hard, so variational inference replaces integration with optimisation: introduce a tractable family $q(z)$ and push it as close to the posterior as possible. The lever is the **evidence lower bound**.

Theorem 2.1 (The ELBO and its gap)**.** For any distribution $q(z)$,

$$\log p(x) = \underbrace{\mathbb{E}_q\left[\log \frac{p(x, z)}{q(z)}\right]}_{\mathrm{ELBO}(q)} + \underbrace{\mathrm{KL}\big(q(z) \,\|\, p(z \mid x)\big)}_{\geq 0}.$$

Hence $\mathrm{ELBO}(q) \leq \log p(x)$, with equality iff $q(z) = p(z \mid x)$.

Proof. Write $p(x, z) = p(z \mid x)p(x)$ inside the ELBO: $\mathbb{E}_q[\log p(z \mid x) + \log p(x) - \log q(z)] = \log p(x) - \mathrm{KL}(q \| p(z \mid x))$, since $\mathbb{E}_q[\log p(x)] = \log p(x)$ and the remaining terms are $-\mathrm{KL}$. Rearranging gives the identity, and $\mathrm{KL} \geq 0$ gives the bound. □

Because $\log p(x)$ is fixed, maximising the ELBO over q *minimises* the KL to the true posterior — the right panel shows the ELBO as a function of a Gaussian q's mean, rising to touch $\log p(x)$ exactly when q hits the posterior, with the KL as the shaded gap. This is the same decomposition that made EM a monotone ascent in Part II, Chapter 6: EM's E-step is the special case where the posterior is tractable and q can be set equal to it; variational inference is what you do when it is not, optimising the ELBO over a restricted family instead.

Example 2.1 (The variational autoencoder)**.** The **VAE** is variational inference with neural networks doing the bookkeeping. An encoder network outputs the parameters of $q(z \mid x)$, a decoder network parameterises $p(x \mid z)$, and the training objective is the ELBO, $\mathbb{E}_q[\log p(x \mid z)] - \mathrm{KL}(q(z \mid x) \,\|\, p(z))$ — a reconstruction term minus a KL to the prior. The ELBO's expectation is estimated by Monte Carlo (Section 1), closing the circle: variational inference turns the integral into an optimisation, and Monte Carlo estimates the one expectation that remains.

Summary

Monte Carlo estimates an intractable integral $\mathbb{E}_p[f]$ by averaging samples: unbiased, with dimension-free standard error $\sigma/\sqrt{n}$; importance sampling reweights draws from a proposal q by $w = p/q$ and is the tool for rare events. Variational inference turns the evidence integral into an optimisation via the ELBO: $\log p(x) = \mathrm{ELBO}(q) + \mathrm{KL}(q \| p(z \mid x))$, so maximising the (tractable) ELBO minimises KL to the posterior. EM is the tractable-posterior special case (Part II Ch. 6); the VAE is the neural case, with an encoder for q, a decoder for $p(x \mid z)$,

and the ELBO — a reconstruction minus a KL — as its loss, its expectation estimated by Monte Carlo.